



$$\mathbf{L} = \mathbf{D} - \mathbf{A} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 3 & -1 & -1 \\ 0 & -1 & 2 & -1 \\ 0 & -1 & -1 & 2 \end{bmatrix}$$

degrees (1, 3, 2, 2), Fiedler vector  $(-2, 0, 1, 1)$ , eigenvalues  $\lambda = (0, 1, 3, 4)$ ;  $\lambda_2 = 1 > 0$ : connected